

ORIGINAL ARTICLE

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Cardiovascular responses to cold-water immersions of the forearm and face, and their relationship to apnoea

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Abstract Apnoea as well as cold stimulation of the face or the extremities elicits marked cardiovascular reflexes in humans. The purpose of this study was to investigate whether forearm immersion in cold water has any effect on the cardiovascular responses to face immersion and apnoea. We recorded cardiovascular responses to cold-water immersions of the forearm and face in 19 (part I) and 23 subjects (part II). The experimental protocol was divided in two parts, each part containing four tests: I₁, forearm immersion during eupnoea; I₂, face immersion during eupnoea; I₃, forearm and face immersion during eupnoea; I₄, face immersion during apnoea; II₁, apnoea without immersion; II₂, forearm immersion during apnoea; II₃, face immersion during apnoea; and II₄, forearm and face immersion during apnoea. The water temperature was 9–11 °C. Cold-water immersion of either the forearm or face was enough to elicit the most pronounced thermoregulatory vasoconstriction during both eupnoea and apnoea. During eupnoea, heart rate responses to forearm immersion (3% increase) and face immersion (9% decrease) were additive during concurrent stimulation (3% decrease). During apnoea, the heart rate responses were not affected by the forearm immersion. The oxygen-conserving diving response seems to dominate over thermoregulatory responses in the threat of asphyxia. During breathing, however, the diving response serves no purpose and does not set thermoregulatory adjustments aside.

Key words Diving response · Bradycardia · Tachycardia · Arterial blood pressure · Cold stimulation

Introduction

Apnoea or cold stimulation of either the face or the extremities elicits marked cardiovascular reflexes in humans. Stimulation of both the sympathetic and parasympathetic nervous systems are involved in evoking these reflexes. The human diving response is initiated by apnoea and is augmented by immersion of the face in cold water (Gooden 1994). It is characterised by a reduction in heart rate, which is caused by an increased parasympathetic activity via the vagus nerve that inhibits the cardiac pacemaker, and an alpha-adrenergic sympathetic vasoconstriction in selective vascular beds (Heistad et al. 1968; Frey et al. 1980a; Gooden 1994). The reduction in cardiac output is not balanced to the vasoconstriction and, as a consequence, blood pressure is increased. The diving response is augmented by stimulation of cold-receptors in the upper facial region, which is innervated by the ophthalmic branch of the trigeminal nerve (Schuitema and Holm 1988).

Cold stimulation of the extremities during eupnoea (the cold pressure test) elicits an inhibition of the parasympathetic activity and increases the beta-adrenergic sympathetic activity to the heart, resulting in a tachycardia. At the same time, an alpha-adrenergic sympathetic vasoconstriction is induced (Frey et al. 1980a, b; Allen et al. 1992). These two responses cause an increase in blood pressure. As the immersion continues, stimulation of the arterial baroreceptors returns the heart rate towards the pre-immersion rate, often within the 1st min (i.e. the beta-adrenergic sympathetic activity is reduced while the alpha-adrenergic sympathetic activity continues to result in vasoconstriction and increased blood pressure; Frey et al. 1980a, b).

According to previous studies, chilling of the forearm and the face has opposite effects on heart rate. The purpose of the study presented here was to investigate whether forearm immersion in cold water has any effect on the cardiovascular responses to face immersion and apnoea (the diving response). This was done by

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recording the cardiovascular responses to cold-water immersion of the forearm and face during eupnoea and apnoea.

Methods

Subjects

A group of 34 healthy subjects, 23 men and 11 women, ranging in age from 19 to 30 years (mean 23.4 years) participated in this study. All subjects were non-smokers but two were taking snuff. Their physical exercise averaged 3.3 h/week, with a range from 0 to 9.5 h/week.

Measurements

An electrocardiogram was checked for anomalies prior to testing (Cardisun 501, lead 2, Fukuda ME Kogoyo, Tokyo, Japan). The subject's vital capacity was measured (Spirolite 201, Vise Medical, Chiba, Japan), first in the standing position and thereafter with the subject lying prone on a bed with the equipment attached. Heart rate and arterial blood pressure were recorded continuously in all subjects on the left middle finger by a photoplethysmometer (Finapres 2300, Ohmeda, Madison, Wisc., USA). Skin capillary blood flow in the left thumb was recorded continuously in 22 of the subjects with a laser Doppler flowmeter (Advanced Laser Flowmeter 21, Advance, Tokyo, Japan). The left hand was positioned at the same level relative to the heart throughout the experiment (cf Schagatay and Andersson 1998). Heart rate, blood pressure, and skin blood flow values were stored continuously in a computer. Breathing movements were recorded by a pneumatic chest bellows, which was connected to an amplifier and an analogue recorder.

Experimental procedures

All experiments were conducted in conformity with the principles of the Declaration of Helsinki and were approved by the Research Ethics Committee of the Faculty of Medicine, Lund University. The experimental protocol was divided into two parts, each part containing four tests as follows. For part I:

1. Eupnoea with forearm immersion.
2. Eupnoea with face immersion.
3. Eupnoea with forearm and face immersion.
4. Apnoea with face immersion.

And for part II:

1. Apnoea without immersion.
2. Apnoea with forearm immersion.
3. Apnoea with face immersion.
4. Apnoea with forearm and face immersion.

Nineteen subjects performed part I [mean (SD) age 23 (3) years, height 1.77 (0.1) m, body mass 70 (12) kg, vital capacity 5.3 (1.0) l], and 23 subjects performed part II [mean (SD) age 24 (2) years, height 1.79 (0.1) m, body mass 72 (10) kg, vital capacity 5.2 (1.0) l]. Eight subjects participated in both parts of the study. The forearm and face were immersed in water at 9–11 °C. Room temperature was 22–26 °C. The order in which the tests were conducted was randomised for each subject, and for the eight subjects that participated in both parts, part I and part II were performed on separate days.

The subjects were familiarised with the equipment and thoroughly informed about the experimental procedures, after which an informed consent form was signed. The subject assumed a prone position on a mattress with the head resting on a removable pillow above the face-immersion trough. The subject's arms were resting on each side of the face-immersion trough. To the right of the bed, the forearm immersion trough was situated on a small table. If the

test did not include forearm immersion, the subject was instructed to lay his/her arm on the table. During forearm immersion tests, the right forearm and hand were immersed. If the test did not include face immersion, the subject was instructed to hold his/her face just over the water. During face immersions, the entire face including the forehead and chin were immersed. This assured an identical position during all the tests. The subject was instructed to breathe normally throughout the experiment and to make a deep, but not maximal inspiration when initiating apnoea. In all of the eupnoea tests the subject started breathing through a snorkel at least 2 min before the beginning of the test. The individual respiratory frequency and tidal volume during the control period before the immersions were continued during immersions as judged by observations of the breathing movements on the analogue recorder.

The recording of all parameters was commenced 2 min prior to each test. The pillow was removed 15 s prior to the tests. At this point, the subject was instructed to put his/her right arm in position beside the trough. A countdown was given during the last 10 s. During apnoea tests, the initiations of apnoea and immersion were simultaneous. Immersions without apnoea lasted for 60 s, while apnoeas were performed to individual maximal duration [mean (SD) for all apnoeas: 108 (49) s]. After the immersion/apnoea was terminated, the recording continued for a further 2 min. The pause between each test comprised 10 min of prone rest.

Data analysis

The control values for heart rate, skin blood flow, and mean arterial pressure were calculated as time averages from the period 90–30 s before each test. The time averages of these parameters were also calculated during the period 30–45 s into each test. The relative changes were calculated between the time averages from the period 30–45 s into the tests and the corresponding control values. We chose the period 30–45 s into the tests for comparative analysis because other studies indicate that the heart rate effect of forearm immersion in cold water is of a short duration (Frey et al. 1980a, b), and because it takes approximately 30 s for the human diving response to develop completely (Jung and Stolle 1981; Andersson and Schagatay 1998b; Schagatay et al. 1999). Thus, the chosen period was considered as the optimum for investigating the interaction between the responses to forearm and face immersion and their relationship to apnoea. For statistical evaluations within the two parts of the study, a paired *t* test with Bonferroni correction for multiple comparisons was used. For comparisons between the two apnoeas with face immersions, an unpaired *t* test was used. The level of statistical significance was set at $P < 0.05$.

Results

Development of cardiovascular responses

Part I

During the forearm immersion, the heart rate increased slightly, but towards the end of the 60-s immersion the heart rate had returned to the control level (Fig. 1). The skin capillary blood flow was reduced throughout the entire forearm immersion. The mean arterial pressure increased during the forearm immersion. During the face immersion while breathing through a snorkel, a similar pattern of cardiovascular responses could be observed, except for the heart rate response. After an initial tachycardic response, a heart rate reduction well below the control level developed. During the concurrent face and forearm immersion, the skin blood flow and blood

pressure followed the same pattern as during forearm or face immersion, while the heart rate response was of an intermediate magnitude compared to during forearm or face immersion. After an initial tachycardic response, apnoea with face immersion induced a definite heart rate reduction. The skin blood flow during apnoea with face immersion followed the same pattern as during the other tests, while the increase in blood pressure was more marked during the apnoea than in any other test in part I.

Part II

During all of the apnoeas, a heart rate reduction followed an initial tachycardic response that started even before the beginning of apnoea (Fig. 2). The heart rate reduc-

tions observed during the apnoeas without face immersions were not as marked as those observed during the apnoeas with face immersions. The skin capillary blood flow was reduced during all of the apnoeas. During the apnoea without immersion, the final level of skin blood flow adjustment was higher than in the other tests. After initial reductions during the first 20 s, the mean arterial pressure increased throughout the apnoeic episodes. Again, the responses during apnoeas without face immersions were not as marked as in the other two tests.

Anticipatory responses

For both heart rate and skin blood flow, anticipatory responses beginning about 30 s prior to apnoeas or

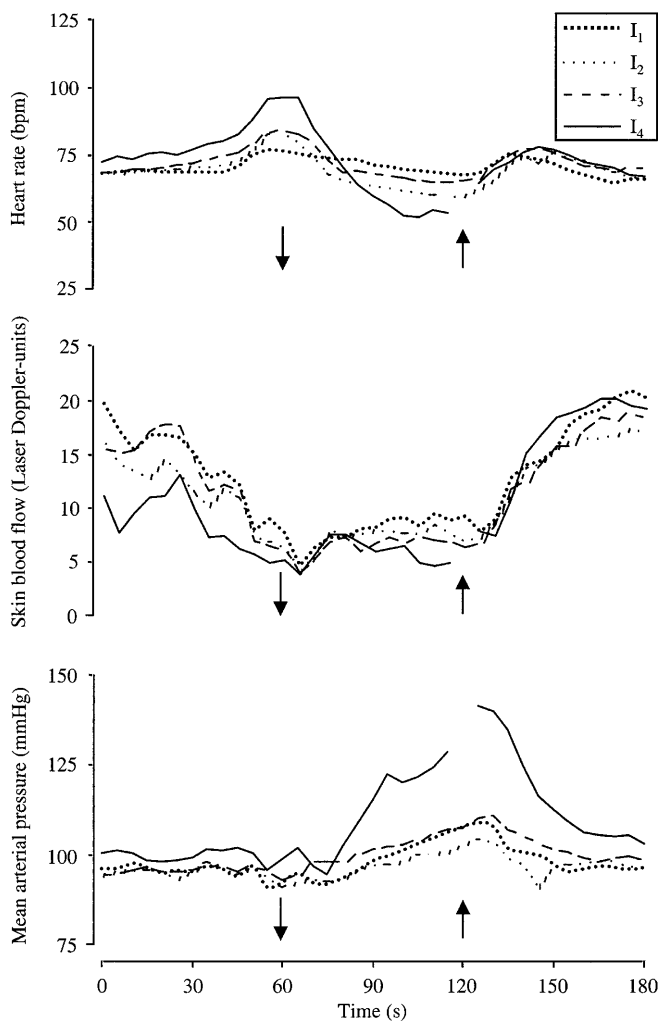


Fig. 1 Mean heart rate, skin capillary blood flow, and mean arterial pressure before, during, and after tests I₁–I₄: I₁ eupnoea with forearm immersion; I₂ eupnoea with face immersion; I₃ eupnoea with face and forearm immersion; I₄ apnoea with face immersion. Heart rate and arterial pressure: $n = 19$; Skin blood flow: $n = 14$. Downward-pointing arrows indicate the start of the tests, and upward-pointing arrows indicate the end of the tests. The break in the line of test I₄ indicates that the apnoea was of a longer duration than 60 s

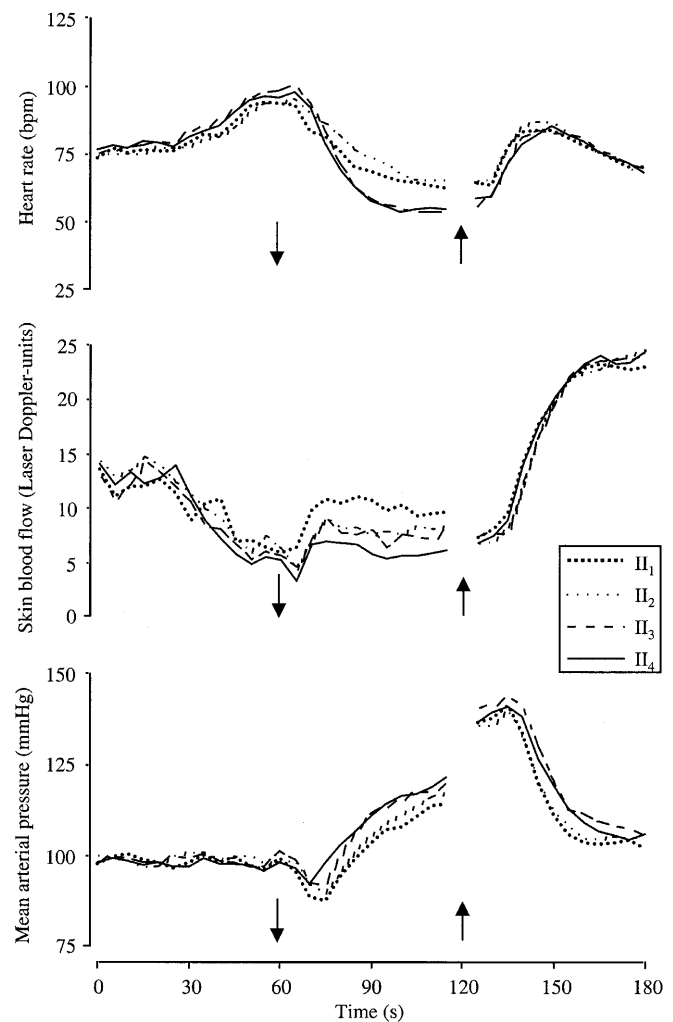


Fig. 2 Mean heart rate, skin capillary blood flow, and mean arterial pressure before, during, and after tests II₁–II₄: II₁ apnoea without immersion; II₂ apnoea with forearm immersion; II₃ apnoea with face immersion; II₄ apnoea with forearm and face immersion. Heart rate and arterial pressure: $n = 23$; Skin blood flow: $n = 16$. Downward-pointing arrows indicate start of the tests, and upward-pointing arrows indicate the end of tests. The break in the lines indicates that the apnoeas were of longer duration than 60 s

immersions could be observed. We performed additional experiments on six subjects to elucidate the effects of “dummy immersions” on the anticipatory responses. With the same experimental set-up, after an identical countdown, subjects had the forearm lowered (by the experimenter) into a trough with or without cold water. Subjects were unaware of the presence or absence of the water until the forearm was lowered into the trough, as confirmed by post-test interviews. Results similar to those observed in the main experiment were observed during the “true immersion” (Fig. 3). The heart rate increased slightly, by 1%, compared to the control level during immersions, and the skin blood flow was reduced by 54%. During the “dummy immersion”, the cardiovascular parameters followed a different pattern (Fig. 3). The heart rate decreased by 4% and the skin blood flow increased by 15%.

Comparisons between cardiovascular responses to immersion and apnoea

There were no differences in the cardiovascular responses to apnoea with face immersion between parts I and II.

Heart rate

The relative changes in heart rate from the control levels during the period 30–45 s into the tests were compared (Fig. 4). Within part I, the apnoea with face immersion resulted in a heart rate response that differed from all the other tests ($P < 0.01$). The forearm immersion and the face immersion resulted in different responses ($P < 0.01$). The responses in these two tests did not differ from the response during concurrent face and

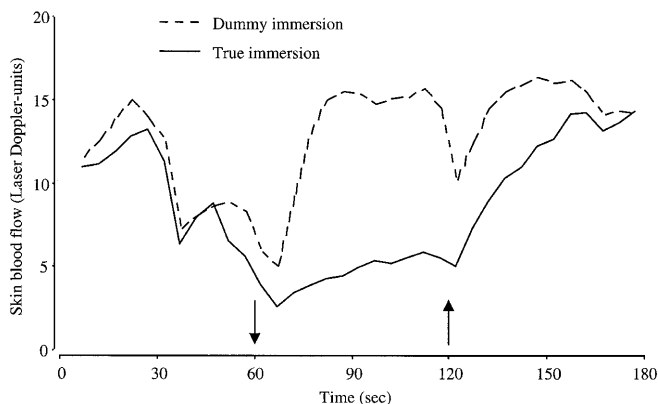


Fig. 3 Mean skin capillary blood flow before, during, and after the forearm was lowered into a trough with (true immersion, solid line) or without cold water (dummy immersion, dashed line). Values are means from six subjects. Downward-pointing arrows indicate start of the immersions, and upward-pointing arrows indicate the end of immersions.

forearm immersion. Within part II, the apnoeas without face immersions resulted in less pronounced heart rate reductions when compared to the apnoeas with face immersions ($P < 0.001$). However, heart rate responses during the apnoea in air and the apnoea with forearm immersion did not differ. Neither did the responses during the apnoea with face immersion and the apnoea with face and forearm immersion differ.

Skin blood flow

The relative changes in skin capillary blood flow did not differ among tests I_{1-4} (Fig. 4). However, the skin blood flow reduction during the apnoea in air was less pronounced than the reductions during the apnoea with forearm immersion and apnoea with face and forearm immersion ($P < 0.05$).

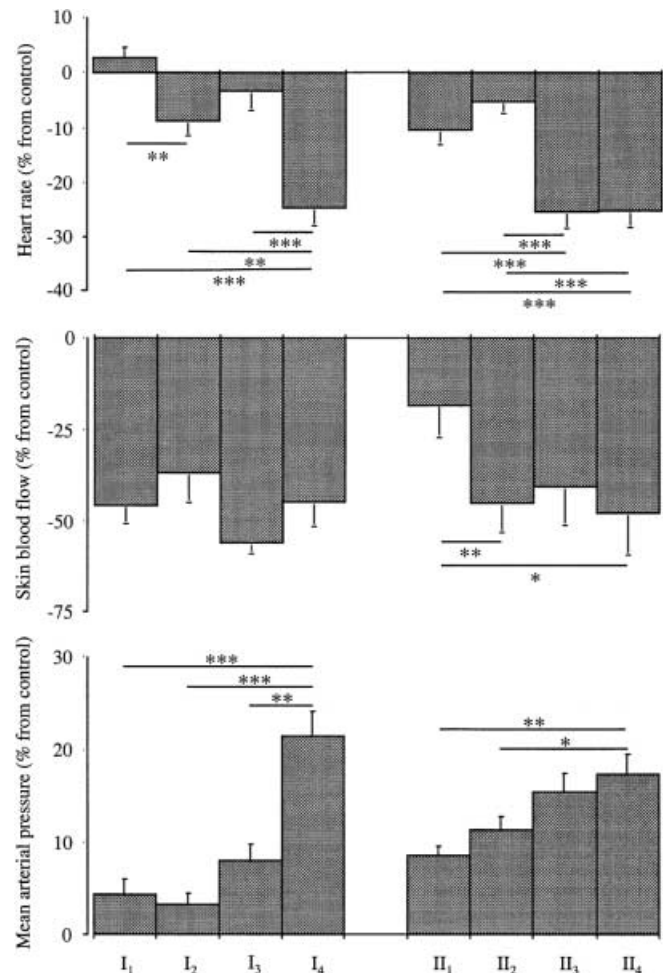


Fig. 4 Percentage changes in heart rate [$n = 19$ (part I) or $n = 23$ (part II)], skin capillary blood flow [$n = 14$ (part I) or $n = 16$ (part II)], and mean arterial pressure [$n = 19$ (part I) or 23 (part II)] from the control period for tests I_1 – I_4 and II_1 – II_4 (see Figs. 1, 2). Data are presented as the means (SEM). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Mean arterial pressure

The mean arterial pressure increased more ($P < 0.01$) during the apnoea than in any other test within part I (Fig. 4). During the apnoea with face and forearm immersion, the mean arterial pressure increased more than in the apnoeas without face immersions ($P < 0.05$).

Discussion

Earlier studies have shown that the cardiovascular responses to cold stimulation of the extremities can be additive to the responses from other cardiovascular challenges (e.g. changes in central blood volume and isometric exercise; Ebert et al. 1982; Peikert and Smolander 1991). There has been, to our knowledge, no other study investigating whether cold stimulation of the extremities affects the cardiovascular responses to apnoea, the human diving response. Our results indicate that during eupnoea, the heart rate response to simultaneous cold stimulation of the forearm and face is an additive response to the two stimuli, while the skin blood flow and arterial pressure responses are not further augmented by the combined stimulation. During apnoea the pattern changed. Instead of an additive response, a complete development of the human diving response occurred (i.e. cold stimulation of the forearm does not attenuate the heart rate reduction, nor does it augment the skin blood flow or blood pressure response).

Cold stimulation of the extremities elicits thermoregulatory reflexes that result in a reduction in heat loss to the environment. One of the most prominent responses is a reduction in peripheral blood flow. As the result of a general sympathetic activation, the heart rate and heart contractility is increased (Frey et al. 1980a). The vasoconstriction and increased cardiac output leads to an increase in the arterial blood pressure. This pattern of cardiovascular reflexes could be observed in the results of this study. Immersion of the face in cold water elicited a different cardiovascular response. Stimulation of the facial cold receptors resulted in a heart rate reduction and a reduced peripheral blood flow. These autonomic reflexes are brought about by vagal chronotropic effects on the heart and alpha-adrenergic sympathetic outflow to resistance vessels (Finley et al. 1979; Frey et al. 1980a; Fagius and Sundlof 1986). Since the reduced cardiac output in this situation was not balanced to the increased total peripheral resistance, the arterial blood pressure was elevated. When the two stimuli (forearm and face immersion) were combined, an additive response pattern was observed regarding the heart rate. The heart rate during concurrent forearm and face immersion was at an intermediate level compared to during the forearm or face immersion. However, the vasoconstriction obtained during either the forearm or face immersion was not further augmented by cold stimulation of both areas simultaneously, as indicated

by the lack of differences in skin capillary blood flow among the tests of part I in this study (Figs. 1, 4).

The heart rate reduction and selective vasoconstriction elicited by apnoea without immersion were augmented by immersion of the face in cold water, as reported previously (Kawakami et al. 1967; Hurwitz and Furedy 1986; Marsh et al. 1995; Andersson and Schagatay 1998a). However, forearm immersion did not affect the cardiovascular responses during apnoea as it did during eupnoea (i.e. the human diving response was unaffected by cold stimulation of the forearm during the apnoeas). When comparing the response to apnoea with that obtained during the apnoea with forearm immersion, we found no difference in heart rate or blood pressure responses. Neither were any differences found when the responses during the apnoea with face immersion and the apnoea with face and forearm immersion were compared. Therefore, we conclude that forearm immersion has no effect on the development of the general diving response when immersed simultaneously as apnoea begins.

The reduction in skin capillary blood flow during apnoea was augmented by cold-water immersion. In fact, all immersions resulted in similar skin capillary blood flow levels, during both eupnoea and apnoea. Cold stimulation is the most effective stimulus causing reflex vasoconstriction in the skin (Heistad and Abboud 1974). Thus, our results indicate that cold stimulation, regardless of the site of stimulation, elicits a vasoconstriction in skin vessels that can not be further augmented by simultaneous cold stimulation of another part of the body surface.

The mean arterial blood pressure was more elevated during the apnoeas than during the immersions with eupnoea. It has been shown that the cardiac output is reduced during breath holding (Ferrigno et al. 1986). The increased blood pressure thus implies that the total peripheral resistance has increased more than the cardiac output decreased. Thermal stimuli primarily affect skin blood flow (Heistad and Abboud 1974). Apnoea also affects blood flow in other vascular beds (e.g. muscle blood flow; Zbrozyna and Westwood 1992). The findings of higher arterial pressures during the apnoeas compared to the eupnoeic immersions could thus be due to a vasoconstriction in vessels not affected by thermal stimuli. At the onset of apnoea, there was a transient reduction in blood pressure. This is caused by the elevated intrathoracic pressure that follows holding the breath with a large lung volume, initially impeding the venous return and consequently reducing stroke volume and cardiac output (Linér 1994; Andersson and Schagatay 1998b).

There were anticipatory responses during the countdown before the tests, but true physiological responses developed during the immersions and apnoeas. Anticipatory responses were also clearly visible before the "dummy immersion", in particular as transient reductions in skin blood flow after the instructions "60 s left" and "30 s left" (Fig. 3). However, when the subjects

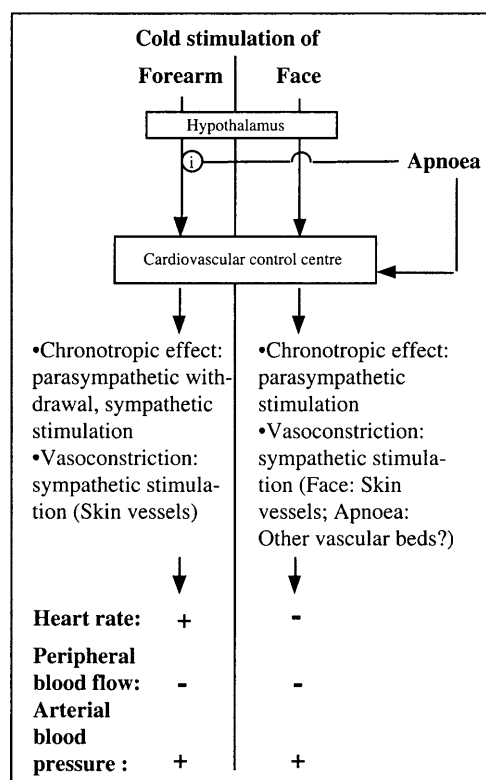


Fig. 5 Schematic model for the integration of the cardiovascular reflexes evoked by the three stimuli (cold stimulation of the forearm, cold stimulation of the face, and apnoea). (i Inhibition, + increase, - decrease)

were surprised by the absence of water, the cardiovascular parameters promptly returned to the pre-count-down level. In particular, the skin blood flow, which also showed the most marked anticipatory responses, increased almost instantaneously towards the control level during the dummy immersion.

Based on the results from this and previous studies, a schematic model for the integration of the cardiovascular reflexes elicited by the three stimuli (cold stimulation of the forearm, cold stimulation of the face, and apnoea) is suggested (Fig. 5). During eupnoea the heart rate reduction derived from chilling of the face is attenuated if the forearm is chilled simultaneously. Thus, the increased chronotropic sympathetic stimulation evoked by arm immersion counteracts the chronotropic parasympathetic activity induced by facial chilling. The peripheral blood flow reduction is not further augmented by cold stimulation of yet another skin area (i.e. the maximal reduction in skin capillary blood flow during immersion in cold water is obtained by either immersion of the face or forearm separately). During apnoea, on the contrary, the forearm immersion has no effect on the heart rate response. Thus, a heart rate reduction was induced by an increased parasympathetic activity, whereas the sympathetic stimulation evoked by forearm immersion was not expressed. The augmented increase in blood pressure during apnoea compared to that during eupnoeic immersions might be explained by a vasoconstriction in specific vascular beds,

besides the reduction in skin blood flow, which would further increase the total peripheral resistance.

The main conclusions from this study are that during eupnoea, heart rate responses from forearm and face immersions are additive, while the heart rate responses to apnoea are not attenuated by the forearm immersion. The fact that the oxygen-conserving diving response has priority over thermoregulatory responses in the threat of asphyxia appears to be a functional property. During breathing, however, the diving response serves no purpose, since oxygen delivery is undisturbed and thermoregulatory adjustments will be more appropriate during cold stimulation.

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